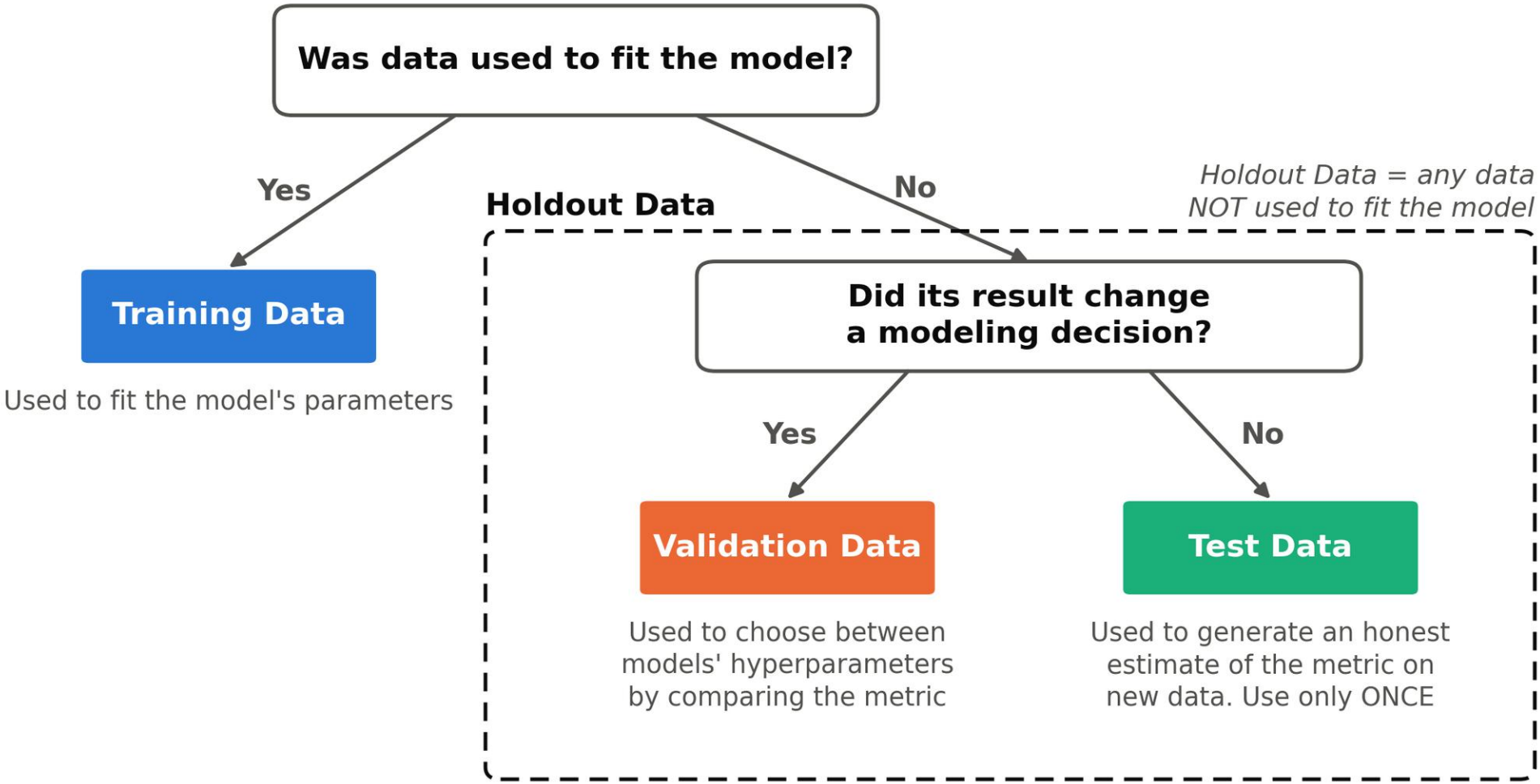
The Data We're Given

	x1	x2	x3	y	
New data (the future)	-0.5	0.2	-0.5	?	**y unknown: what we actually want to predict**
	-0.1	0.3	0.7	?	
	0.2	-0.4	0.3	?	
Data we're given	-0.6	-1.0	-0.2	7.7	**y known**
	-0.3	-0.4	-0.5	4.3	
	0.3	-0.3	1.5	2.3	
	1.4	0.7	-0.4	2.8	
	-1.6	0.8	-0.9	6.0	
	-0.6	-0.2	0.6	3.6	
	-0.3	1.9	-1.1	1.8	
	-0.9	0.9	1.1	1.3	
	-0.3	-1.2	-1.1	6.2	
	1.0	2.2	0.6	2.2	

Data Set Terminology & Purposes

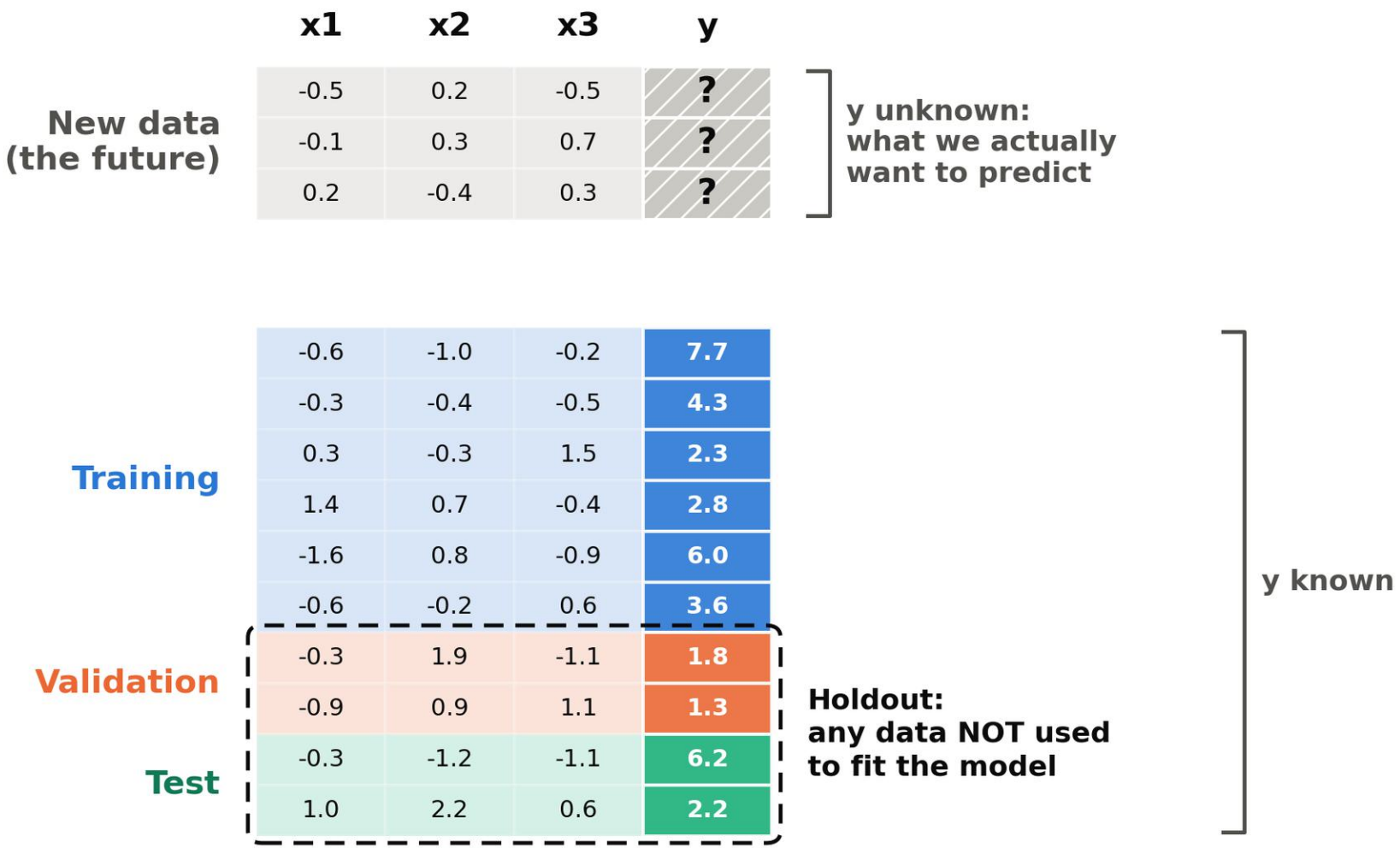
Say you've chosen a metric/score to evaluate the performance of your model's predictions (e.g. RMSE, accuracy)



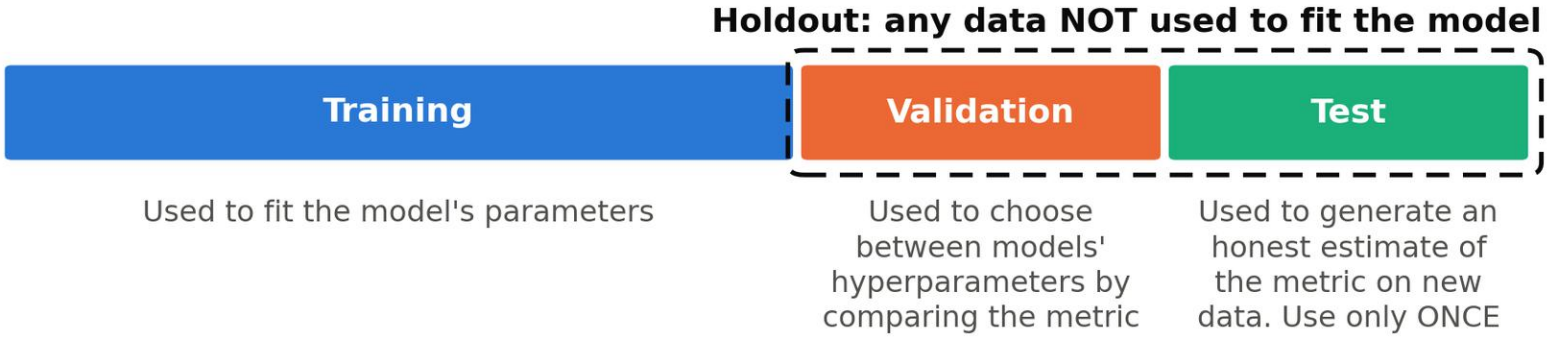
Lock the Test Set Away

	x1	x2	x3	y	
New data (the future)	-0.5	0.2	-0.5	?	**y unknown: what we actually want to predict**
	-0.1	0.3	0.7	?	
	0.2	-0.4	0.3	?	
Rest of the data	-0.6	-1.0	-0.2	7.7	**y known**
	-0.3	-0.4	-0.5	4.3	
	0.3	-0.3	1.5	2.3	
	1.4	0.7	-0.4	2.8	
	-1.6	0.8	-0.9	6.0	
	-0.6	-0.2	0.6	3.6	
	-0.3	1.9	-1.1	1.8	
	-0.9	0.9	1.1	1.3	
Test	-0.3	-1.2	-1.1	6.2	**Locked away until the end**
	1.0	2.2	0.6	2.2	

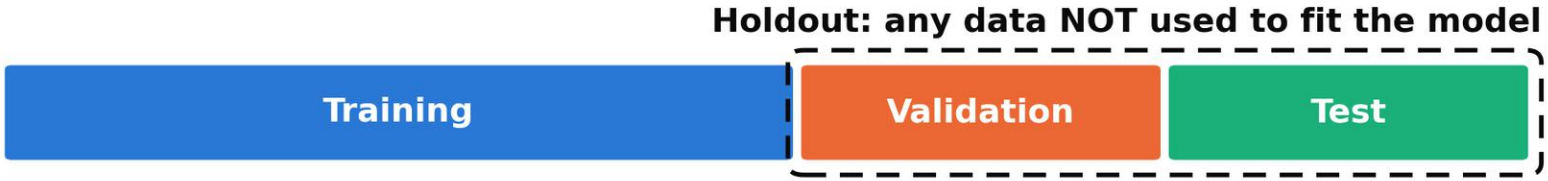
Split the Rest into Training & Validation



Validation set approach: Split once → Get 1 score per model



Validation set approach: Split once → Get 1 score per model



5-fold cross-validation: Rotate the validation fold → Get 5 scores per model, average them



Overall Workflow

1. Lock the test set away
before looking at anything



2. Cross-validate on the rest
pick the hyperparameter value with the best CV metric (e.g. RMSE)



→ Best hyperparameter value

fold models are thrown away

3. Refit the chosen model
on ALL non-test data



→ final model

4. Unlock the test set ONCE
honest estimate of the metric on new data



→ $\widehat{\text{RMSE}} = 2.3$

5. Deploy: predict new data
where y is truly unknown



→ $\hat{y}$, expect $\widehat{\text{RMSE}} \approx 2.3$

Summary: Training vs. Validation vs. Test

Holdout: any data NOT used to fit the model

	Training	Validation	Test
Used to fit the model?	**Yes**	**No**	**No**
Used to make modeling decisions? (e.g. set hyperparameters)	**No**	**Yes**	**No**
When do we look at it?	Throughout fitting	Repeatedly, while tuning	Only ONCE, only AFTER all decisions are made
Purpose	Used to fit the model's parameters	Used to choose between models' hyperparameters by comparing the metric	Used to generate an honest estimate of the metric on new data. Use only ONCE
Analogy	Homework problems	Practice exams	The real final exam